Cayleb Alvarez-James

AI Agent Systems Engineer - Autonomous Agents, Agent Harnesses & Self-Improving Infrastructure

Middletown, Connecticut, USA | caylebalvarezjames@gmail.com

github.com/cayleb-james2008 | linkedin.com/in/cayleb-alvarez-james-99850531b | Portfolio

Summary

AI agent systems engineer and technical founder. I have spent the past year on agentic systems and harness engineering, starting while I was finishing high school, where I was also wrestling team captain for two years. For most of that stretch I worked 13 to 14 hours a day on it, around school, my job and wrestling. I work across several harnesses instead of just one: OpenCode and DeepSeek Harness, Claude Desktop and Duet, Gemini Antigravity, Freebuff, ChatGPT Codex and Claude Code. I pick whichever has usage left and fits the job. Right now most of the work runs through Muse, Claude Duet and the DSH Fleet I am finishing. Three systems make up most of it. Ergon's employee agents set themselves up: the agent watches how someone works, asks them targeted questions, learns the role and keeps a memory of the person, adding skills and tools as it goes. The DSH Fleet is a distributed orchestration layer: it runs dynamic swarms across desktop, virtual machine and browser surfaces, lets outside agents join through Slack, and improves its own workflows, skills and plugin behaviour from the results of finished jobs. OpenBell is my trading research system: I set the weights of tiny models directly, freeze their predictions before scoring, and score them in time order with costs counted. 1,200+ GitHub contributions in the last year. 4,000+ automated tests across five shipped systems.

Experience

Founder - Independent AI Agent Systems | Oct 2025-present

The current builds:

Selected Projects

Ergon (Node.js, C#, Python, SQLite)

- Self-configuring agent that learns each employee's role on its own. In development; 622 tests pass.

DSH Fleet (Python, TypeScript)

- Distributed, self-improving infrastructure: dynamic swarms plus the Omni Loop.

OpenBell (Python, Rust)

- Weight-space training: directly authored tiny specialist models, frozen before scoring.

Sophos (React, TypeScript, Tauri)

- Coding-agent desktop app; sessions survive restarts. 1,224 interface tests; merged after full CI.

dotz (Rust, Tauri, JavaScript, SQLite)

- Runs and supervises AI coding agents. 828+ core tests per platform; all 6 hosted jobs pass; merged.

Asmodeus (Rust)

- Tests trading ideas on past data in time order; no peeking, no real money. 1,111 Rust tests pass.

Industry AI Suite (Python)

- Ten local review tools; every finding shows its source and a person decides. 382 Python tests pass.

Skills

Multi-harness agent operations | dynamic-swarm orchestration | the Omni Loop

Python | TypeScript | JavaScript | Rust | weight-space models | React | Tauri | SQLite

Additional Experience

Robert H. Lord Company (RHL) - AI Concept Discussion & Warehouse-Board Demo | Sep 2026-present

David Cooke Plaster Co. - Plaster Work | Summer 2026

Dunkin' | 2024-2026

Tommy's Pizza | Summer 2024

Education

Portland High School | Portland, Connecticut | Graduated 2026

Coursework: Cybersecurity; AP Computer Science